CISUC
CENTRE FOR INFORMATICS AND SYSTEMS OF THE UNIVERSITY OF COIMBRA

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BONDER
A LIGHTWEIGHT GNN FOR BOND PERCEPTION UNDER NOISE

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1. The Bottleneck

Most AI-driven generative chemistry [1] and some molecular dynamics simulators' outputs consist of raw atomic 3D coordinates. Downstream tasks require structured chemical graphs (an actual molecule), and therefore a conversion step must be taken.

2. Traditional Heuristics and Their Flaws

Traditional tools (e.g., RDKit [2], OpenBabel [3]) rely on rigid interatomic distance thresholds. As coordinate noise increases — a common artifact in generative AI — these rules often fail, yielding shattered or chemically invalid molecules, and silently breaking sensitive properties like aromaticity. Furthermore, they discard vital contextual information, such as neighborhood geometry.

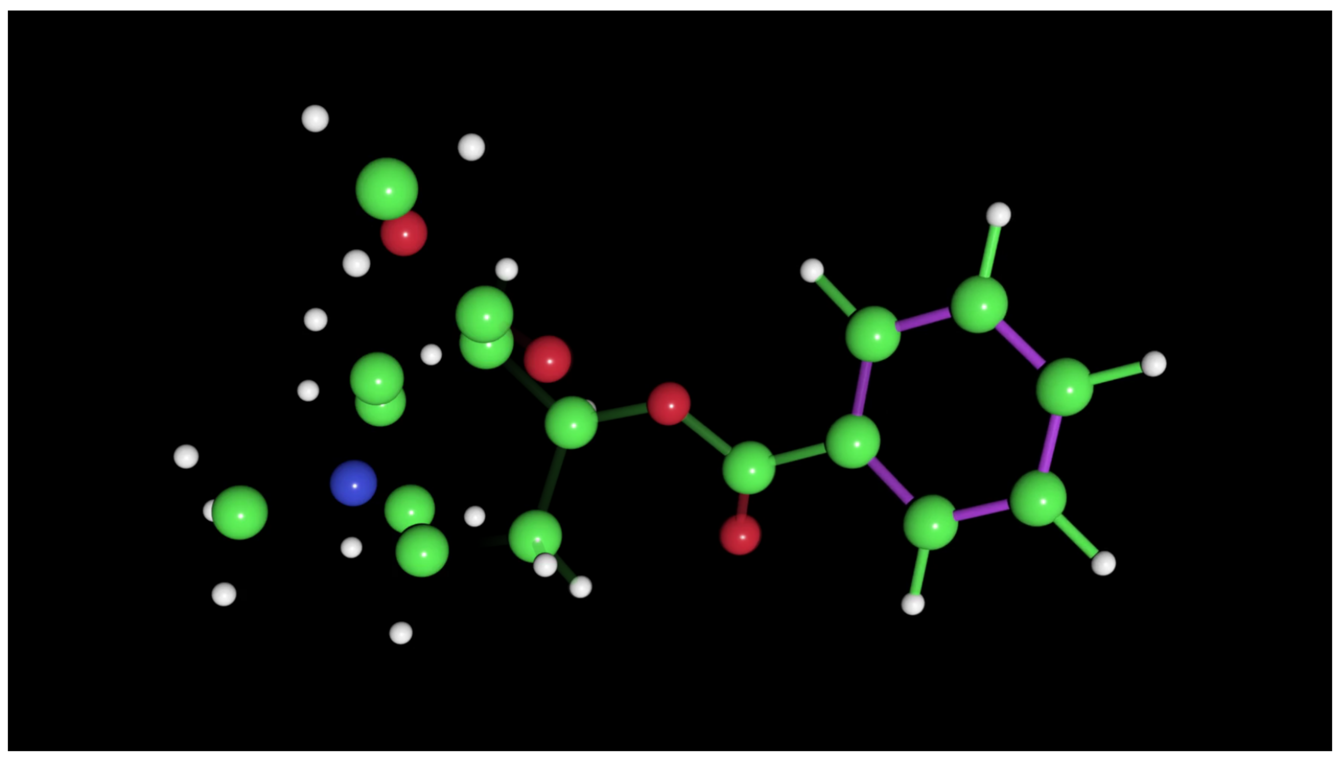

Figure 1: Bonder allows you to perceive bonds in a disconnected atom cloud.

3. Bonder: A Robust, Invariant Solution

Bonder replaces brittle distance cutoffs with a noise-tolerant, continuous deep learning architecture. It seamlessly fuses multiple sources of information, combining standard interatomic distances with learned local topological priors. Local geometry is baked into the continuous message-passing network, allowing the model to gracefully resolve coordinate noise while predicting five distinct bond classes concurrently.

Engineered for practicality, it is a lightweight, drop-in replacement for traditional heuristics. Its architecture is strictly E(3) invariant by design. Furthermore, its compact formulation enables efficient CPU inference, preventing bottlenecks on critical GPU hardware.

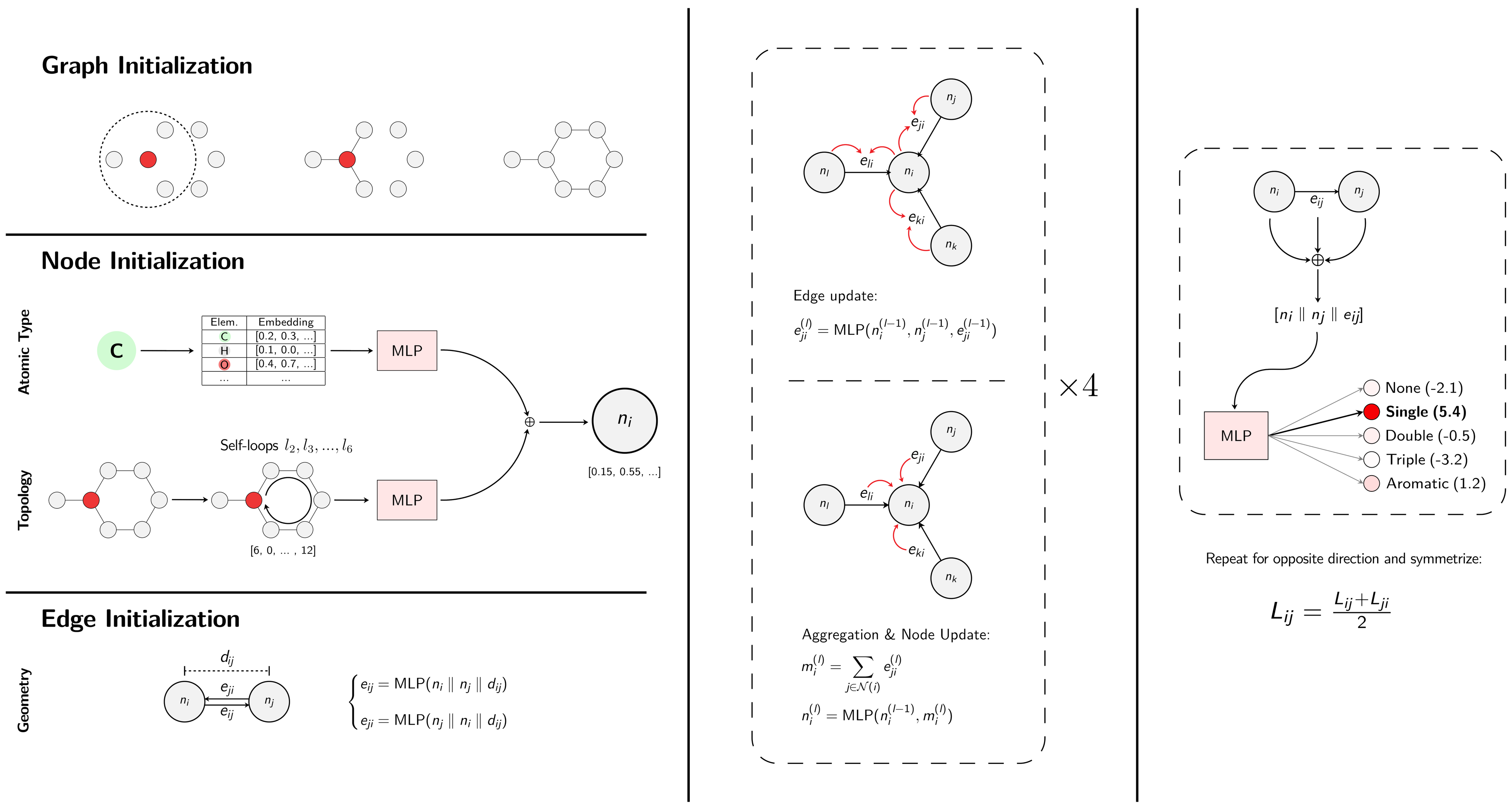

Figure 2: Architecture of Bonder, from feature extraction and network initialization to message-passing mechanics and final prediction head.

4. Initialization & Native E(3) Invariance

The network is initialized as a radius graph. Node representations fuse three distinct information streams: **periodic** (atomic element embeddings), **topological** (self-loops extracted from adjacency matrix powers A^2 through A^6), and **geometric** (continuous pairwise distances). By strictly processing relative pairwise distances rather than absolute 3D coordinates, the architecture guarantees native E(3) invariance.

5. Directional, Edge-First Message Passing

Operating over a 4-layer dense network, the continuous message-passing loop is structured to update edge states before node states. However, edges are treated directionally — maintaining independent representations for $i \rightarrow j$ and $j \rightarrow i$. This asymmetry incorporates and transmits the different contexts on each side of the edge.

6. Optimization and Symmetrization

To improve edge representational capacity, the loss function is evaluated directly on the final directional logits. However, to respect the physical reality that chemical bonds are undirected, the paired directional logits are averaged post-loss, yielding a single non-directional prediction.

7. Results

Bonder matches baseline accuracy at zero noise ($< 0.5\%$ accuracy difference) strictly from raw coordinates, while still bypassing the need for *a priori* formal charges, provided to the base methods. Under Gaussian noise, baselines degrade rapidly, while Bonder maintains near-maximum accuracy. While still limited to logit-level evaluation, this serves as a rigorous proxy for future full reconstruction.

8. Limitations & Deployment

While bond predictions are accurate, outputting 100% chemically valid graphs (e.g., SDFs) requires formal charge assignment. We are actively integrating this directly into the dense message-passing loop to become a fully standalone drop-in replacement for traditional methods.
For deployment, Bonder will launch both as a Python package and as a serverless WebAssembly web app (via ONNXRuntime).

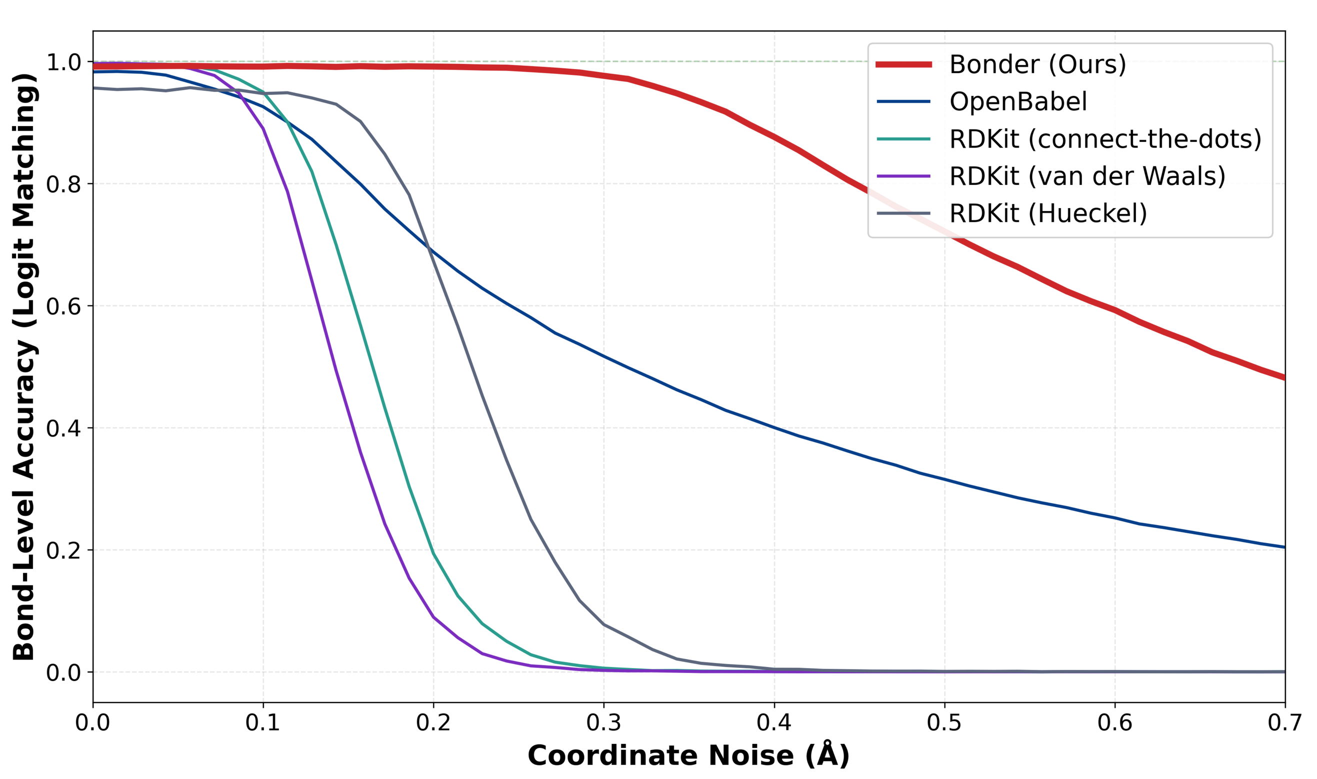

Figure 3: Bond-level accuracy when noise is progressively injected. Bonder maintains near perfect stability even with extreme noise corruption.

Acknowledgments

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References

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